

Complex Numbers

Definitions

1. Define a complex number. Provide an example.
2. Explain the imaginary unit i and its properties.
3. Define the real and imaginary parts of a complex number.
4. Explain the concept of the complex conjugate. Provide an example.
5. Define a purely real and purely imaginary complex number. Give examples.
6. Explain what it means for a complex number to be in standard form.
7. Define the modulus (magnitude) of a complex number. Give an example.
8. Explain the argument of a complex number. Provide an example.
9. Define the polar form of a complex number. Give an example.
10. Explain the rectangular form of a complex number. Provide an example.

Operations

1. Perform the addition of the complex numbers $2 + 3i$ and $-4 - 2i$.
2. Calculate the product of the complex numbers $1 + 2i$ and $3 - 4i$.
3. Perform the division of the complex numbers $\frac{5+2i}{1-i}$.
4. Compute the square of the complex number $2 + i$.
5. Simplify the expression $(1 + i)^3$ using De Moivre's theorem.
6. Compute the cube root of the complex number -8 using polar form.
7. Find the reciprocal of the complex number $2 - 3i$.
8. Calculate the modulus of the complex number $4 + 3i$.
9. Compute the argument of the complex number $-1 - \sqrt{3}i$.
10. Perform the multiplication of the complex numbers in polar form: $r_1(\cos \theta_1 + i \sin \theta_1)$ and $r_2(\cos \theta_2 + i \sin \theta_2)$.

Polar Form and De Moivre's Theorem

Polar Form

1. Express the complex number $3 + 4i$ in polar form.
2. Convert the polar form $r(\cos \theta + i \sin \theta)$ to rectangular form.
3. Given the polar form $2(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$, find the rectangular form.
4. Express the complex number $-6 - 8i$ in polar form.
5. Convert the polar form $5(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2})$ to rectangular form.

De Moivre's Theorem

1. State and explain De Moivre's Theorem.
2. Use De Moivre's Theorem to find the fourth power of the complex number $1 + i$.
3. Given the complex number $2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$, use De Moivre's Theorem to find its fifth power.
4. Express $(1 - i)^6$ in rectangular form using De Moivre's Theorem.
5. Find the square root of the complex number $4(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$ using De Moivre's Theorem.

Practical Problems

1. Represent the solution to the quadratic equation $x^2 + 1 = 0$ in terms of complex numbers.
2. A vector in the complex plane is represented by the complex number $2 + 3i$. Find its magnitude.
3. Solve the equation $z^2 + 1 = 0$ for z in terms of complex numbers.
4. Express the solutions to the equation $z^3 = -8$ in polar form.
5. A current in an electrical circuit is represented by the complex number $I = 5\angle 30^\circ$. Find the real and imaginary components of the current.